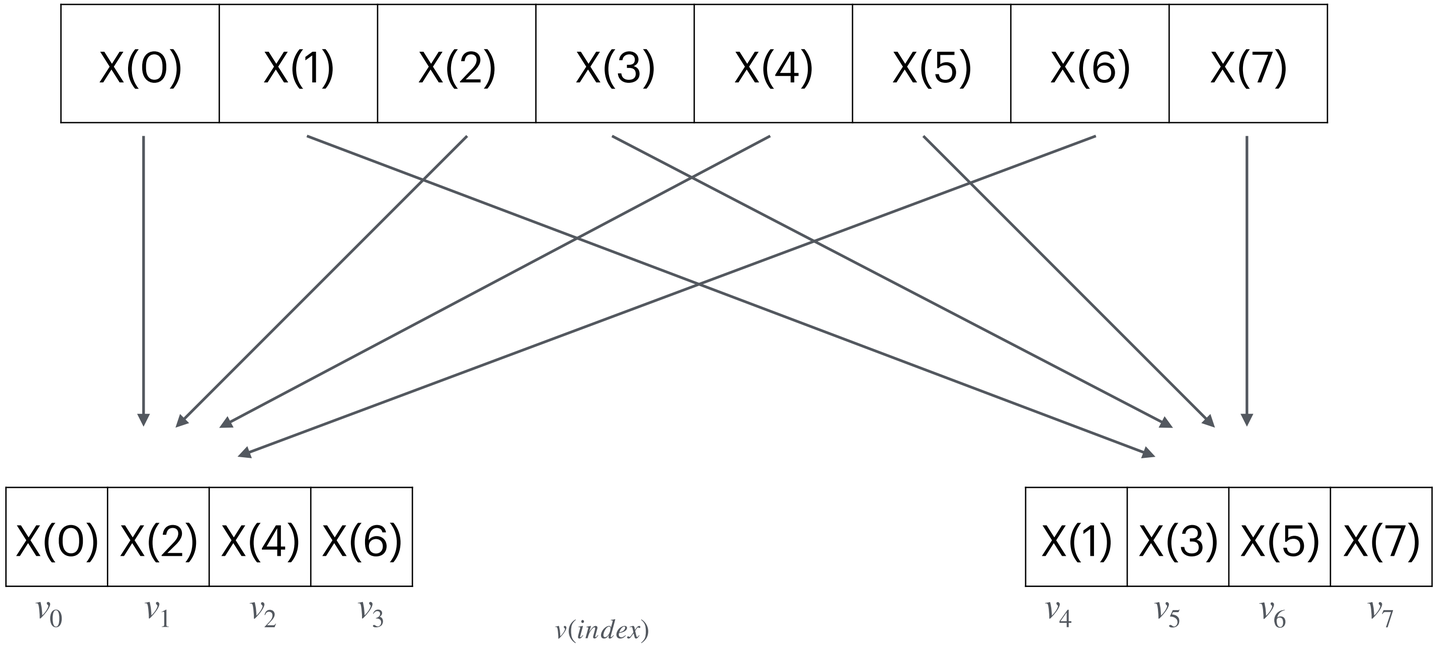
Radix FFT

Decimation in Time

$$X(m) = \sum_{n=0}^{N-1} x(n) e^{\frac{-j2\pi nm}{N}}$$



$$X(m) = \sum_{n=0}^{(N/2)-1} \overset{x(index)}{x(2n)} e^{\frac{-j2\pi(2n)m}{N}} \overset{v(index)}{} + \sum_{n=0}^{(N/2)-1} \overset{x(index)}{x(2n+1)} e^{\frac{-j2\pi(2n+1)m}{N}} \overset{v(index)}{}$$

$$X(m) = \sum_{n=0}^{(N/2)-1} x(2n) e^{\frac{-j2\pi(2n)m}{N}} + e^{\frac{-j2\pi m}{N}} \sum_{n=0}^{(N/2)-1} x(2n+1) e^{\frac{-j2\pi 2nm}{N}}$$

$$X(m) = \sum_{n=0}^{(N/2)-1} x(2n)e^{\frac{-j2\pi(2n)m}{N}} \quad + \quad e^{\frac{-j2\pi m}{N}} \sum_{n=0}^{(N/2)-1} x(2n+1)e^{\frac{-j2\pi 2nm}{N}}$$

$$X(m+N/2) = \sum_{n=0}^{(N/2)-1} x(2n)e^{\frac{-j2\pi(2n)(m+N/2)}{N}} \quad + \quad e^{\frac{-j2\pi(m+N/2)}{N}} \sum_{n=0}^{(N/2)-1} x(2n+1)e^{\frac{-j2\pi 2n(m+N/2)}{N}}$$

$$X(m+N/2) = \sum_{n=0}^{(N/2)-1} x(2n)e^{\frac{-j2\pi(2nm+2nN/2)}{N}} \quad + \quad e^{\frac{-j2\pi m)}{N}} e^{\frac{-j2\pi N/2)}{N}} \sum_{n=0}^{(N/2)-1} x(2n+1)e^{\frac{-j2\pi 2n(m+N/2)}{N}}$$

$$X(m+N/2) = \sum_{n=0}^{(N/2)-1} x(2n)e^{\frac{-j2\pi(2nm)}{N}} e^{\frac{-j2\pi(2nN/2)}{N}} \quad + \quad e^{\frac{-j2\pi m)}{N}} e^{\frac{-j2\pi N/2)}{N}} \sum_{n=0}^{(N/2)-1} x(2n+1)e^{\frac{-j2\pi 2nm}{N}} e^{\frac{-j2\pi 2n(N/2)}{N}}$$

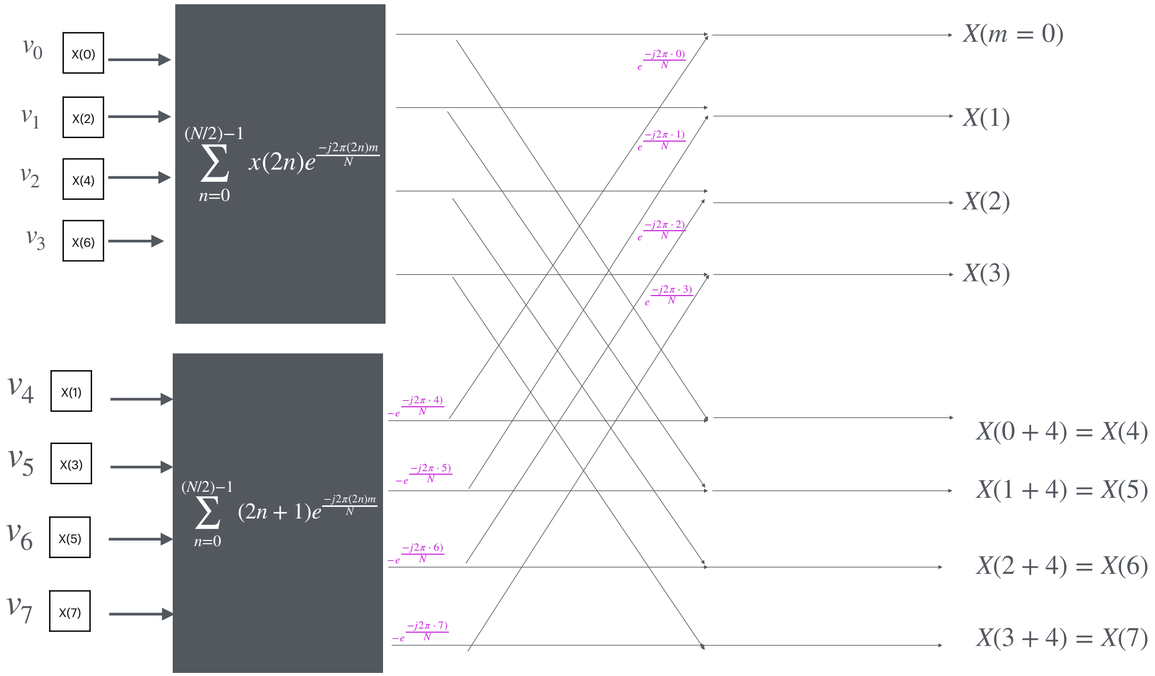
$$X(m+N/2) = \sum_{n=0}^{(N/2)-1} x(2n)e^{\frac{-j2\pi(2nm)}{N}} e^{\frac{-j2\pi(nN)}{N}} \quad + \quad e^{\frac{-j2\pi m}{N}} e^{\frac{-j\pi N}{N}} \sum_{n=0}^{(N/2)-1} x(2n+1)e^{\frac{-j2\pi 2nm}{N}} e^{\frac{-j2\pi n}{}}$$

$$X(m+N/2) = \sum_{n=0}^{(N/2)-1} x(2n)e^{\frac{-j2\pi(2nm)}{N}} e^{\frac{-j2\pi(n)}{}} \quad + \quad e^{\frac{-j2\pi m}{N}} e^{\frac{-j\pi}{}} \sum_{n=0}^{(N/2)-1} x(2n+1)e^{\frac{-j2\pi 2nm}{N}} 1$$

$$X(m+N/2) = \sum_{n=0}^{(N/2)-1} x(2n)e^{\frac{-j2\pi(2nm)}{N}} \quad + \quad -e^{\frac{-j2\pi m}{N}} \sum_{n=0}^{(N/2)-1} x(2n+1)e^{\frac{-j2\pi 2nm}{N}}$$

X(0)	X(2)	X(4)	X(6)
v_0	v_1	v_2	v_3

X(1)	X(3)	X(5)	X(7)
v_4	v_5	v_6	v_7



$$X(m) = \sum_{n=0}^{(N/2)-1} x(2n)e^{\frac{-j2\pi(2n)m}{N}} \quad + \quad e^{\frac{-j2\pi m}{N}} \sum_{n=0}^{(N/2)-1} x(2n+1)e^{\frac{-j2\pi 2nm}{N}}$$

$$X(m+N/2) = \sum_{n=0}^{(N/2)-1} x(2n)e^{\frac{-j2\pi(2nm)}{N}} \quad + \quad -e^{\frac{-j2\pi m}{N}} \sum_{n=0}^{(N/2)-1} x(2n+1)e^{\frac{-j2\pi 2nm}{N}}$$

X(0)	X(2)	X(4)	X(6)
v_0	v_1	v_2	v_3

$$\sum_{n=0}^{(N/2)-1} x(2n)e^{\frac{-j2\pi(2n)m}{N}}$$

X(0)	X(4)
u_0	u_1

X(2)	X(6)
u_2	u_4

$$= \sum_{n=0}^{(N/4)-1} x(4n)e^{\frac{-j2\pi 2(2n)m}{N}} + \sum_{n=0}^{(N/4)-1} x(4n+2)e^{\frac{-j2\pi 2(2n+1)m}{N}}$$

$$+ \sum_{n=0}^{(N/4)-1} x(4n+2)e^{\frac{-j2\pi 4nm}{N}} e^{\frac{-j2\pi 2m}{N}}$$

$$A(X) = \sum_{n=0}^{(N/4)-1} x(4n)e^{\frac{-j2\pi(4n)m}{N}} + e^{\frac{-j2\pi 2m}{N}} \sum_{n=0}^{(N/4)-1} x(4n+2)e^{\frac{-j2\pi 4nm}{N}}$$

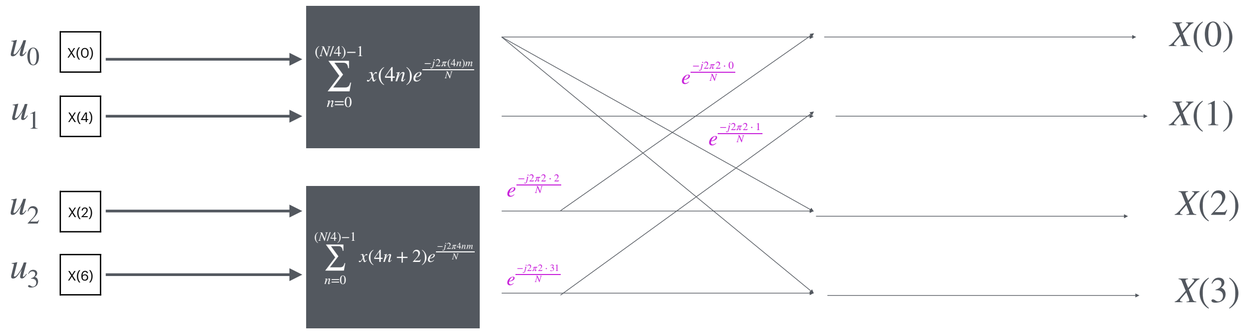
$$X(m) = \sum_{n=0}^{(N/4)-1} x(4n)e^{\frac{-j2\pi(4n)m}{N}} \quad + \quad e^{\frac{-j2\pi 2m}{N}} \sum_{n=0}^{(N/4)-1} x(4n+2)e^{\frac{-j2\pi 4nm}{N}}$$

$$X(m+N/2) = \sum_{n=0}^{(N/4)-1} x(4n)e^{\frac{-j2\pi(4n)(m+N/2)}{N}} \quad + \quad e^{\frac{-j2\pi 2(m+N/2)}{N}} \sum_{n=0}^{(N/4)-1} x(4n+2)e^{\frac{-j2\pi 4n(m+N/2)}{N}}$$

$$X(m+N/2) = \sum_{n=0}^{(N/4)-1} x(4n)e^{\frac{-j2\pi(4n)(m)}{N}} e^{\frac{-j2\pi(4n)N}{2N}} \quad + \quad e^{\frac{-j2\pi 2m}{N}} e^{\frac{-j2\pi 2N}{2N}} \sum_{n=0}^{(N/4)-1} x(4n+2)e^{\frac{-j2\pi 4n(m)}{N}} e^{\frac{-j2\pi 4n(N)}{2N}}$$

$$X(m+N/2) = \sum_{n=0}^{(N/4)-1} x(4n)e^{\frac{-j2\pi(4n)(m)}{N}} e^{\frac{-j2\pi(2n)}{}} \quad + \quad e^{\frac{-j2\pi 2m}{N}} e^{\frac{-j2\pi}{}} \sum_{n=0}^{(N/4)-1} x(4n+2)e^{\frac{-j2\pi 4nm}{N}} e^{\frac{-j2\pi 2n}{}}$$

$$X(m+N/2) = \sum_{n=0}^{(N/4)-1} x(4n)e^{\frac{-j2\pi(4n)(m)}{N}} \quad + \quad e^{\frac{-j2\pi 2m}{N}} \sum_{n=0}^{(N/4)-1} x(4n+2)e^{\frac{-j2\pi 4nm}{N}}$$



$$X(m) = \sum_{n=0}^{(N/4)-1} x(4n)e^{\frac{-j2\pi(4n)m}{N}} + e^{\frac{-j2\pi2m}{N}} \sum_{n=0}^{(N/4)-1} x(4n+2)e^{\frac{-j2\pi4nm}{N}}$$

$$X(m+N/2) = \sum_{n=0}^{(N/4)-1} x(4n)e^{\frac{-j2\pi(4n)(m)}{N}} + e^{\frac{-j2\pi2m}{N}} \sum_{n=0}^{(N/4)-1} x(4n+2)e^{\frac{-j2\pi4nm}{N}}$$

X(1)	X(3)	X(5)	X(7)
v_4	v_5	v_6	v_7

$$\sum_{n=0}^{(N/2)-1} x(2n+1)e^{\frac{-j2\pi 2nm}{N}}$$

X(1)	X(5)
u_0	u_1

X(3)	X(7)
u_2	u_3

$$\sum_{n=0}^{(N/4)-1} x(2(2n)+1)e^{\frac{-j2\pi 2(2n)m}{N}} \quad + \quad \sum_{n=0}^{(N/4)-1} x(2(2n+1)+1)e^{\frac{-j2\pi 2(2n+1)m}{N}}$$

$$\sum_{n=0}^{(N/4)-1} x(4n+1)e^{\frac{-j2\pi 4nm}{N}} \quad + \quad \sum_{n=0}^{(N/4)-1} x((4n+2)+1)e^{\frac{-j2\pi 2(2n+1)m}{N}}$$

$$\sum_{n=0}^{(N/4)-1} x(4n+1)e^{\frac{-j2\pi 4nm}{N}} \quad + \quad \sum_{n=0}^{(N/4)-1} x(4n+3)e^{\frac{-j2\pi 2(2n)m}{N}} e^{\frac{-j2\pi 2(1)m}{N}}$$

$$B(X) = \sum_{n=0}^{(N/4)-1} x(4n+1)e^{\frac{-j2\pi 4nm}{N}} \quad + \quad e^{\frac{-j2\pi 2m}{N}} \sum_{n=0}^{(N/4)-1} x(4n+3)e^{\frac{-j2\pi 4nm}{N}}$$

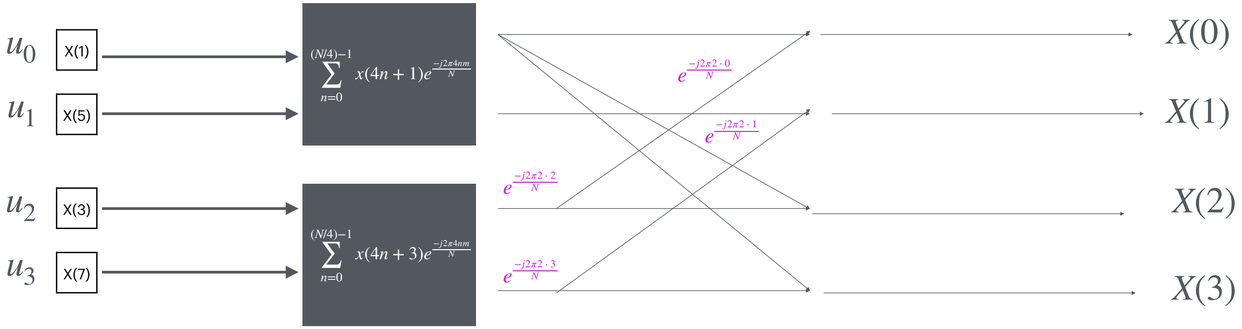
$$B(X+N/2) = \sum_{n=0}^{(N/4)-1} x(4n+1)e^{\frac{-j2\pi 4n(m+N/2)}{N}} \quad + \quad e^{\frac{-j2\pi 2(m+N/2)}{N}} \sum_{n=0}^{(N/4)-1} x(4n+3)e^{\frac{-j2\pi 4n(m+N/2)}{N}}$$

$$B(X+N/2) = \sum_{n=0}^{(N/4)-1} x(4n+1)e^{\frac{-j2\pi 4n(m)}{N}} e^{\frac{-j2\pi 4n(N/2)}{N}} \quad + \quad e^{\frac{-j2\pi 2(m)}{N}} e^{\frac{-j2\pi 2(N/2)}{N}} \sum_{n=0}^{(N/4)-1} x(4n+3)e^{\frac{-j2\pi 4n(m)}{N}} e^{\frac{-j2\pi 4n(N/2)}{N}}$$

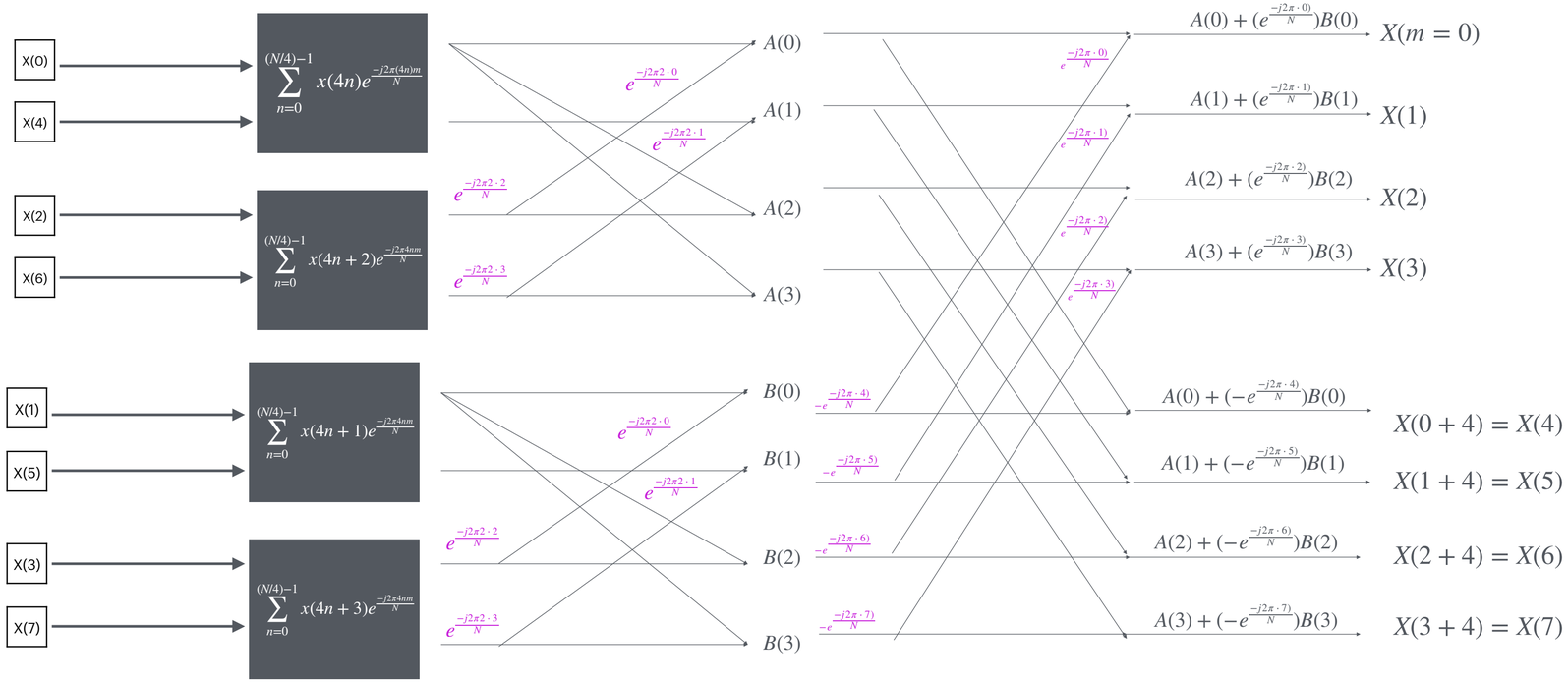
$$B(X+N/2) = \sum_{n=0}^{(N/4)-1} x(4n+1)e^{\frac{-j2\pi 4n(m)}{N}} e^{\frac{-j2\pi 2n}{}} \quad + \quad e^{\frac{-j2\pi 2(m)}{N}} e^{\frac{-j2\pi}{}} \sum_{n=0}^{(N/4)-1} x(4n+3)e^{\frac{-j2\pi 4n(m)}{N}} e^{\frac{-j2\pi 2n()}{}}$$

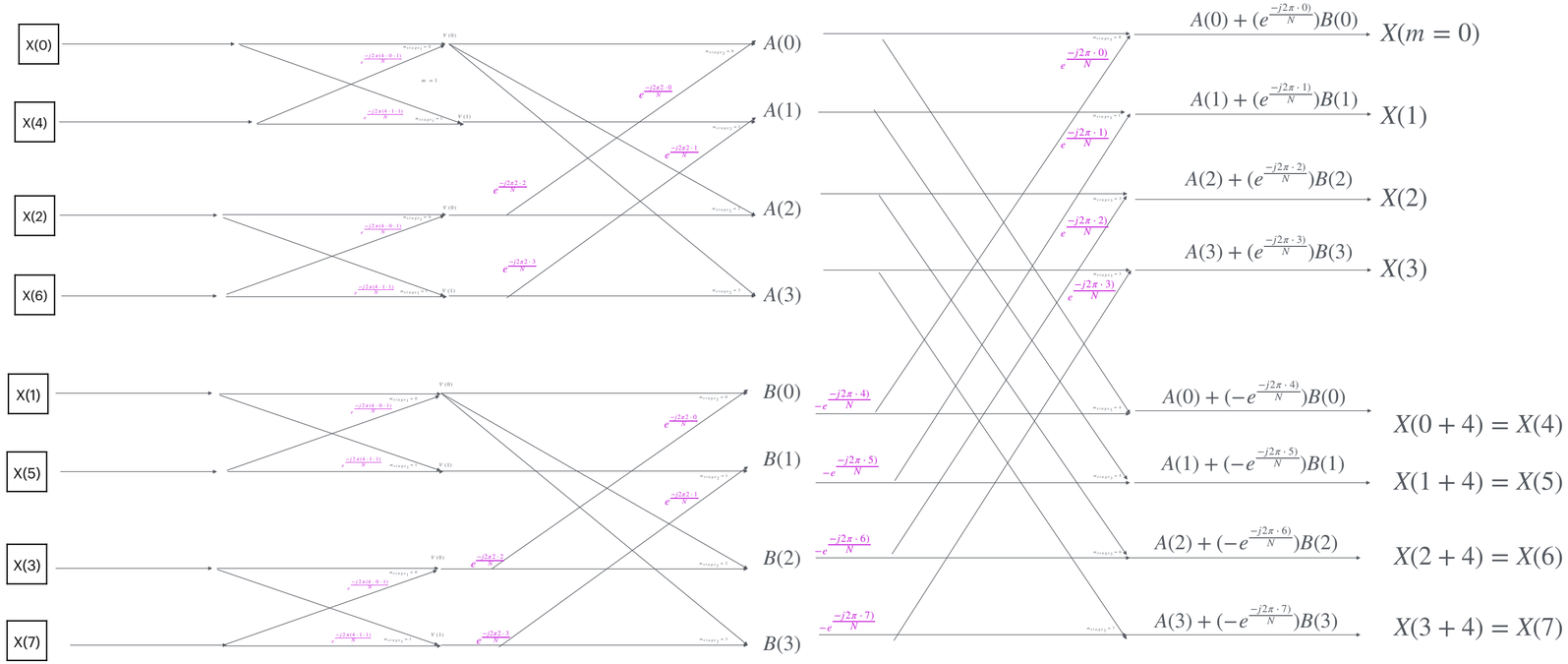
$$B(X+N/2) = \sum_{n=0}^{(N/4)-1} x(4n+1)e^{\frac{-j2\pi 4nm}{N}} \quad + \quad e^{\frac{-j2\pi 2m}{N}} \sum_{n=0}^{(N/4)-1} x(4n+3)e^{\frac{-j2\pi 4nm}{N}}$$

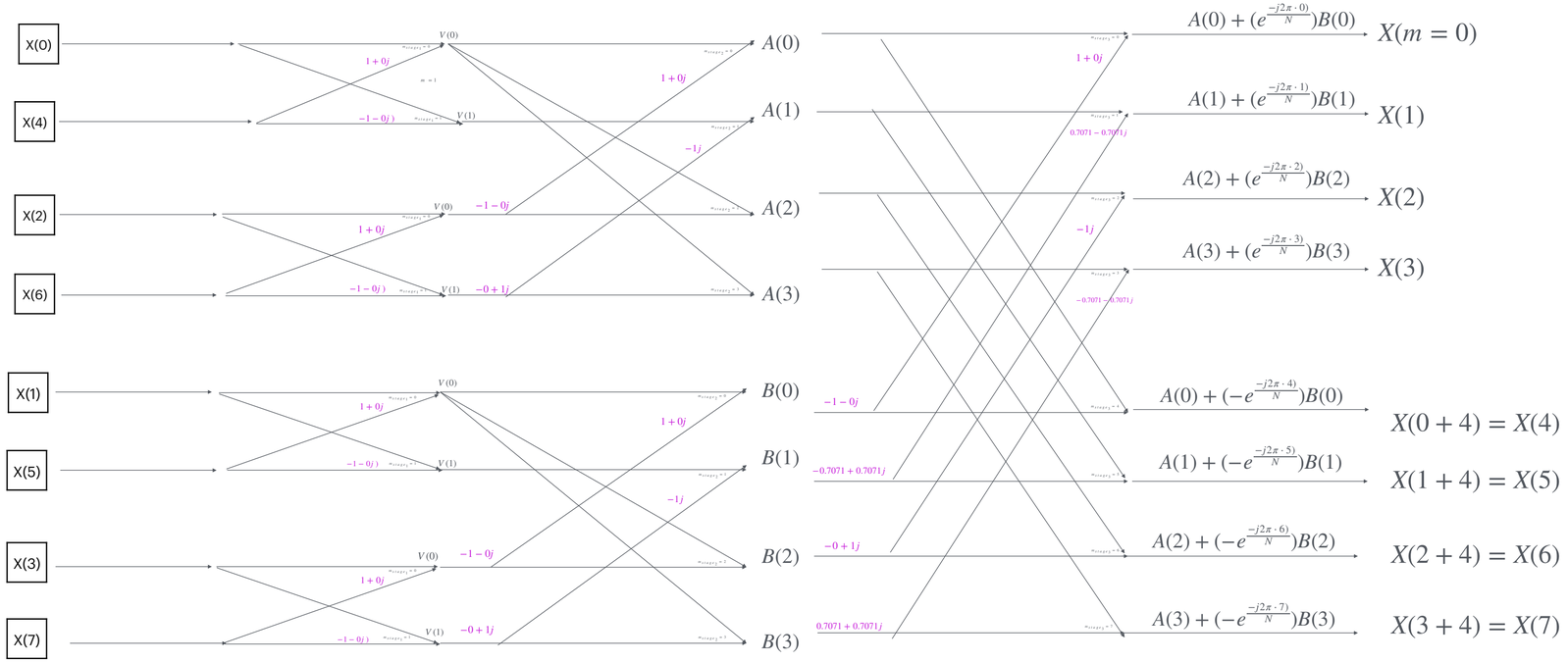
$$B(X+N/2) = B(X)$$



$$B(X) = \sum_{n=0}^{(N/4)-1} x(4n+1)e^{\frac{-j2\pi 4nm}{N}} \quad + \quad e^{\frac{-j2\pi 2m}{N}} \sum_{n=0}^{(N/4)-1} x(4n+3)e^{\frac{-j2\pi 4nm}{N}}$$


X(0)
X(4)
X(2)
X(6)
X(1)
X(5)
X(3)
X(7)
$\sum_{n=0}^{(N/4)-1} x(4n)e^{\frac{-j2\pi(4n)m}{N}}$
$\sum_{n=0}^{(N/4)-1} x(4n+2)e^{\frac{-j2\pi 4nm}{N}}$
$\sum_{n=0}^{(N/4)-1} x(4n+1)e^{\frac{-j2\pi 4nm}{N}}$
$\sum_{n=0}^{(N/4)-1} x(4n+3)e^{\frac{-j2\pi 4nm}{N}}$
$e^{\frac{-j2\pi 2\cdot 0}{N}}$
$e^{\frac{-j2\pi 2\cdot 1}{N}}$
$e^{\frac{-j2\pi 2\cdot 2}{N}}$
$e^{\frac{-j2\pi 2\cdot 3}{N}}$
$A(0)$
$A(1)$
$A(2)$
$A(3)$
$B(0)$
$B(1)$
$B(2)$
$B(3)$
$-e^{\frac{-j2\pi\cdot 4)}{N}}$
$-e^{\frac{-j2\pi\cdot 5)}{N}}$
$-e^{\frac{-j2\pi\cdot 6)}{N}}$
$-e^{\frac{-j2\pi\cdot 7)}{N}}$
$e^{\frac{-j2\pi\cdot 0)}{N}}$
$e^{\frac{-j2\pi\cdot 1)}{N}}$
$e^{\frac{-j2\pi\cdot 2)}{N}}$
$e^{\frac{-j2\pi\cdot 3)}{N}}$
$A(0)+(e^{\frac{-j2\pi\cdot 0)}{N}})B(0)$
$A(1)+(e^{\frac{-j2\pi\cdot 1)}{N}})B(1)$
$A(2)+(e^{\frac{-j2\pi\cdot 2)}{N}})B(2)$
$A(3)+(e^{\frac{-j2\pi\cdot 3)}{N}})B(3)$
$A(0)+(-e^{\frac{-j2\pi\cdot 4)}{N}})B(0)$
$A(1)+(-e^{\frac{-j2\pi\cdot 5)}{N}})B(1)$
$A(2)+(-e^{\frac{-j2\pi\cdot 6)}{N}})B(2)$
$A(3)+(-e^{\frac{-j2\pi\cdot 7)}{N}})B(3)$
$X(m=0)$
$X(1)$
$X(2)$
$X(3)$
$X(0+4)=X(4)$
$X(1+4)=X(5)$
$X(2+4)=X(6)$
$X(3+4)=X(7)$


X(0)
X(4)
X(2)
X(6)
X(1)
X(5)
X(3)
X(7)
A(0)
A(1)
A(2)
A(3)
B(0)
B(1)
B(2)
B(3)
$A(0) + (e^{\frac{-j2\pi \cdot 0)}{N}})B(0)$
$A(1) + (e^{\frac{-j2\pi \cdot 1)}{N}})B(1)$
$A(2) + (e^{\frac{-j2\pi \cdot 2)}{N}})B(2)$
$A(3) + (e^{\frac{-j2\pi \cdot 3)}{N}})B(3)$
$A(0) + (-e^{\frac{-j2\pi \cdot 4)}{N}})B(0)$
$A(1) + (-e^{\frac{-j2\pi \cdot 5)}{N}})B(1)$
$A(2) + (-e^{\frac{-j2\pi \cdot 6)}{N}})B(2)$
$A(3) + (-e^{\frac{-j2\pi \cdot 7)}{N}})B(3)$
$X(m = 0)$
$X(1)$
$X(2)$
$X(3)$
$X(0 + 4) = X(4)$
$X(1 + 4) = X(5)$
$X(2 + 4) = X(6)$
$X(3 + 4) = X(7)$


X(0)
X(4)
X(2)
X(6)
X(1)
X(5)
X(3)
X(7)
V(0)
V(1)
1 + 0j
−1 − 0j
−0 + 1j
A(0)
A(1)
A(2)
A(3)
B(0)
B(1)
B(2)
B(3)
0.7071 + 0.7071j
−0.7071 + 0.7071j
0.7071 − 0.7071j
−0.7071 − 0.7071j
$A(0) + (e^{\frac{-j2\pi \cdot 0)}{N}})B(0)$
$A(1) + (e^{\frac{-j2\pi \cdot 1)}{N}})B(1)$
$A(2) + (e^{\frac{-j2\pi \cdot 2)}{N}})B(2)$
$A(3) + (e^{\frac{-j2\pi \cdot 3)}{N}})B(3)$
$A(0) + (-e^{\frac{-j2\pi \cdot 4)}{N}})B(0)$
$A(1) + (-e^{\frac{-j2\pi \cdot 5)}{N}})B(1)$
$A(2) + (-e^{\frac{-j2\pi \cdot 6)}{N}})B(2)$
$A(3) + (-e^{\frac{-j2\pi \cdot 7)}{N}})B(3)$
$X(m = 0)$
$X(1)$
$X(2)$
$X(3)$
$X(0 + 4) = X(4)$
$X(1 + 4) = X(5)$
$X(2 + 4) = X(6)$
$X(3 + 4) = X(7)$

Decimation in Frequency

$$X(m) = \sum_{n=0}^{N-1} x(n) e^{\frac{e^{-j2\pi nm}}{N}}$$

X(0)	X(1)	X(2)	X(3)	X(4)	X(5)	X(6)	X(7)

X(0)	X(1)	X(2)	X(3)

X(4)	X(5)	X(6)	X(7)

$$X(m) = \sum_{n=0}^{\frac{N}{2}-1} x(n) e^{\frac{e^{-j2\pi nm}}{N}} + \sum_{n=0}^{\frac{N}{2}-1} x(n+\frac{N}{2}) e^{\frac{e^{-j2\pi m(n+\frac{N}{2})}}{N}}$$

$$X(m) = \sum_{n=0}^{\frac{N}{2}-1} x(n) e^{\frac{e^{-j2\pi nm}}{N}} + \sum_{n=0}^{\frac{N}{2}-1} x(n+\frac{N}{2}) e^{\frac{-j2\pi mn}{N}} e^{\frac{-j2\pi m\frac{N}{2}}{N}}$$

$$+ \sum_{n=0}^{\frac{N}{2}-1} x(n+\frac{N}{2}) e^{\frac{-j2\pi mn}{N}} e^{\frac{-j2\pi mN}{2N}}$$

$$+ e^{\frac{-j\pi m}{}} \sum_{n=0}^{\frac{N}{2}-1} x(n+\frac{N}{2}) e^{\frac{-j2\pi mn}{N}}$$

$$X(m) = \sum_{n=0}^{\frac{N}{2}-1} x(n) e^{\frac{e^{-j2\pi nm}}{N}} + (-1^m) \sum_{n=0}^{\frac{N}{2}-1} x(n+\frac{N}{2}) e^{\frac{-j2\pi mn}{N}}$$

$$X(m) = \sum_{n=0}^{\frac{N}{2}-1} x(n) e^{\frac{e^{-j2\pi nm}}{N}} \quad + \quad (-1^{m}) \sum_{n=0}^{\frac{N}{2}-1} x(n + \frac{N}{2}) e^{\frac{-j2\pi mn}{N}}$$

Even Samples Split [,0,2,4,6]

$$X(2m) = \sum_{n=0}^{\frac{N}{2}-1} x(n) e^{\frac{e^{-j2\pi n2m}}{N}} \quad + \quad (-1^{2m}) \sum_{n=0}^{\frac{N}{2}-1} x(n + \frac{N}{2}) e^{\frac{-j2\pi 2mn}{N}}$$

$$X(2m) = \sum_{n=0}^{\frac{N}{2}-1} x(n) e^{\frac{e^{-j2\pi n2m}}{N}} \quad + \quad ((-1^{2})^{m}) \sum_{n=0}^{\frac{N}{2}-1} x(n + \frac{N}{2}) e^{\frac{-j2\pi 2mn}{N}}$$

$$X(2m) = \sum_{n=0}^{\frac{N}{2}-1} x(n) e^{\frac{e^{-j2\pi n2m}}{N}} \quad + \quad \sum_{n=0}^{\frac{N}{2}-1} x(n + \frac{N}{2}) e^{\frac{-j2\pi 2mn}{N}}$$

$$X(m) = \sum_{n=0}^{\frac{N}{2}-1} x(n) e^{\frac{e^{-j2\pi nm}}{N}} \quad + \quad (-1^{m}) \sum_{n=0}^{\frac{N}{2}-1} x(n + \frac{N}{2}) e^{\frac{-j2\pi 2mn}{N}}$$

Odd Samples Split [,1,3,5,7]

$$X(2m+1) = \sum_{n=0}^{\frac{N}{2}-1} x(n) e^{\frac{e^{-j2\pi n(2m+1)}}{N}} \quad + \quad (-1^{(2m+1)}) \sum_{n=0}^{\frac{N}{2}-1} x(n + \frac{N}{2}) e^{\frac{-j2\pi(2m+1)n}{N}}$$

$$X(2m+1) = \sum_{n=0}^{\frac{N}{2}-1} x(n) e^{\frac{-j2\pi n(2m)}{N}} e^{\frac{-j2\pi n(1)}{N}} \quad + \quad (-1^{(2m+1)}) \sum_{n=0}^{\frac{N}{2}-1} x(n + \frac{N}{2}) e^{\frac{-j2\pi(2m)n}{N}} e^{\frac{-j2\pi(1)n}{N}}$$

$$X(2m+1) = ((\sum_{n=0}^{\frac{N}{2}-1} x(n) e^{\frac{-j2\pi n(2m)}{N}} \quad + \quad (-1^{(2m+1)}) \sum_{n=0}^{\frac{N}{2}-1} x(n + \frac{N}{2})) e^{\frac{-j2\pi(1)n}{N}}) e^{\frac{-j2\pi(2m)n}{N}}$$

$$X(2m) = (\sum_{n=0}^{\frac{N}{2}-1} x(n) \quad + \quad \sum_{n=0}^{\frac{N}{2}-1} x(n+\frac{N}{2}))e^{\frac{-j2\pi 2mn}{N}}$$

$$X(2m) = \sum_{n=0}^{\frac{N}{2}-1} (x(n) + x(n+\frac{N}{2}))e^{\frac{-j2\pi 2mn}{N}}$$

Even Samples Split [,0,2,4,6]

$$X(2m+1) = ((\sum_{n=0}^{\frac{N}{2}-1} x(n)e^{\frac{-j2\pi n(2m)}{N}} \quad + \quad (-1^{(2m+1)}) \sum_{n=0}^{\frac{N}{2}-1} x(n+\frac{N}{2}))e^{\frac{-j2\pi(1)n}{N}})e^{\frac{-j2\pi(2m)n}{N}}$$

$$X(2m+1) = \sum_{n=0}^{\frac{N}{2}-1} ((x(n) + (-1^{(2m+1)})x(n+\frac{N}{2}))e^{\frac{-j2\pi(1)n}{N}})e^{\frac{-j2\pi(2m)n}{N}}$$

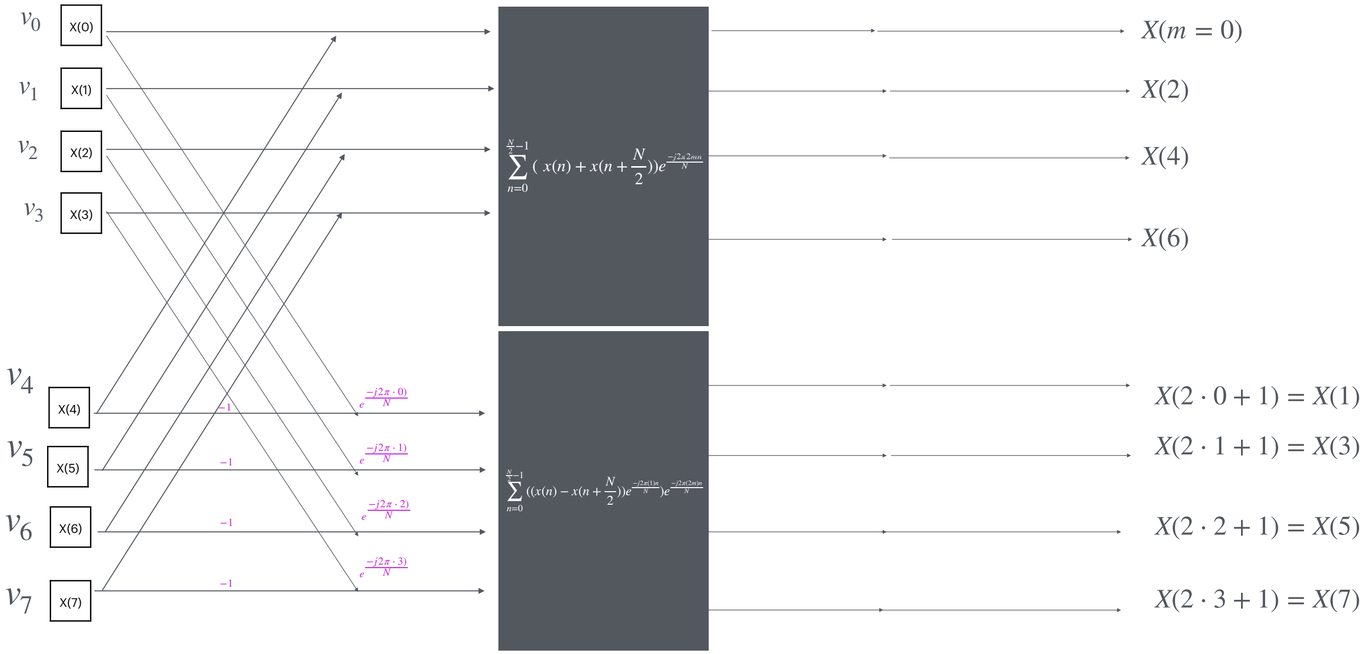
$$X(2m+1) = \sum_{n=0}^{\frac{N}{2}-1} ((x(n) + ((-1^2)^m(-1^1))x(n+\frac{N}{2}))e^{\frac{-j2\pi(1)n}{N}})e^{\frac{-j2\pi(2m)n}{N}}$$

$$X(2m+1) = \sum_{n=0}^{\frac{N}{2}-1} ((x(n) - x(n+\frac{N}{2}))e^{\frac{-j2\pi(1)n}{N}})e^{\frac{-j2\pi(2m)n}{N}}$$

Odd Samples Split [,1,3,5,7]

$$X(2m) = \sum_{n=0}^{\frac{N}{2}-1} \left(x(n) + x(n + \frac{N}{2}) \right) e^{\frac{-j2\pi 2mn}{N}}$$

$$X(2m+1) = \sum_{n=0}^{\frac{N}{2}-1} \left((x(n) - x(n + \frac{N}{2})) e^{\frac{-j2\pi(1)n}{N}} \right) e^{\frac{-j2\pi(2m)n}{N}}$$



$$X(2m) = \sum_{n=0}^{\frac{N}{2}-1} (\ x(n) + x(n+\frac{N}{2}) \ )e^{\frac{-j2\pi 2mn}{N}}$$

$$X(2m) = \sum_{n=0}^{\frac{N}{4}-1} (\ x(n) + x(n+\frac{N}{2}) \ )e^{\frac{-j4\pi mn}{N}} \ + \sum_{n=0}^{\frac{N}{4}-1} (\ x(n+\frac{N}{4}) + x(n+\frac{N}{2}+\frac{N}{4}) \ )e^{\frac{-j4\pi m(n+\frac{N}{4})}{N}}$$

$$X(2m) = \sum_{n=0}^{\frac{N}{4}-1} (\ x(n) + x(n+\frac{N}{2}) \ )e^{\frac{-j4\pi mn}{N}} \ + \sum_{n=0}^{\frac{N}{4}-1} (\ x(n+\frac{N}{4}) + x(n+\frac{N}{2}+\frac{N}{4}) \ )e^{\frac{-j4\pi m(n}{N}}e^{\frac{-j4\pi m(\frac{N}{4})}{N}}$$

$$X(2m) = \sum_{n=0}^{\frac{N}{4}-1} (\ x(n) + x(n+\frac{N}{2}) \ )e^{\frac{-j4\pi mn}{N}} \ + \sum_{n=0}^{\frac{N}{4}-1} (\ x(n+\frac{N}{4}) + x(n+\frac{N}{2}+\frac{N}{4}) \ )e^{\frac{-j4\pi m(n}{N}}e^{\frac{-j4\pi m(\frac{N}{4})}{N}}$$

$$X(2m) = \sum_{n=0}^{\frac{N}{4}-1} (\ x(n) + x(n+\frac{N}{2}) \ )e^{\frac{-j4\pi m(n}{N}} \ + \sum_{n=0}^{\frac{N}{4}-1} (\ x(n+\frac{N}{4}) + x(n+\frac{N}{2}+\frac{N}{4}) \ )e^{\frac{-j4\pi m(n)}{N}}e^{\frac{-j\pi m}{}}$$

$$X(2(2m)) = \sum_{n=0}^{\frac{N}{4}-1} (\ x(n) + x(n+\frac{N}{2}) \ )e^{\frac{-j4\pi(2m)(n)}{N}} \ + \sum_{n=0}^{\frac{N}{4}-1} (\ x(n+\frac{N}{4}) + x(n+\frac{N}{2}+\frac{N}{4}) \ )e^{\frac{-j4\pi(2m)(n)}{N}}e^{\frac{-j\pi(2m)}{}}$$

$$X(4m) = \sum_{n=0}^{\frac{N}{4}-1} (\ x(n) + x(n+\frac{N}{2}) \ )e^{\frac{-j4\pi(2m)(n)}{N}} \ + \sum_{n=0}^{\frac{N}{4}-1} (\ x(n+\frac{N}{4}) + x(n+\frac{N}{2}+\frac{N}{4}) \ )e^{\frac{-j4\pi(2m)(n)}{N}}e^{\frac{-j\pi(2m)}{}}$$

$$X(4m) = \sum_{n=0}^{\frac{N}{4}-1} (\ x(n) + x(n+\frac{N}{2}) \ )e^{\frac{-j2\pi(4m)(n)}{N}} \ + \sum_{n=0}^{\frac{N}{4}-1} (\ x(n+\frac{N}{4}) + x(n+\frac{N}{2}+\frac{N}{4}) \ )e^{\frac{-j2\pi(4m)(n)}{N}}$$

$$X(2m) = \sum_{n=0}^{\frac{N}{4}-1} (\ x(n) + x(n+\frac{N}{2})\ )e^{\frac{-j4\pi m(n}{N}} + \sum_{n=0}^{\frac{N}{4}-1} (\ x(n+\frac{N}{4}) + x(n+\frac{N}{2}+\frac{N}{4})\ )e^{\frac{-j4\pi m(n)}{N}}e^{\frac{-j\pi m}{}}$$

$$X(2(2m+1)) = \sum_{n=0}^{\frac{N}{4}-1} (\ x(n) + x(n+\frac{N}{2})\ )e^{\frac{-j4\pi(2m+1)(n}{N}} + \sum_{n=0}^{\frac{N}{4}-1} (\ x(n+\frac{N}{4}) + x(n+\frac{N}{2}+\frac{N}{4})\ )e^{\frac{-j4\pi(2m+1)(n)}{N}}e^{\frac{-j\pi(2m+1)}{}}$$

$$X((4m+2)) = \sum_{n=0}^{\frac{N}{4}-1} (\ x(n) + x(n+\frac{N}{2})\ )e^{\frac{-j4\pi(2m)(n}{N}}e^{\frac{-j4\pi(1)(n}{N}} + \sum_{n=0}^{\frac{N}{4}-1} (\ x(n+\frac{N}{4}) + x(n+\frac{N}{2}+\frac{N}{4})\ )e^{\frac{-j4\pi(2m)(n)}{N}}e^{\frac{-j4\pi(1)(n)}{N}}e^{\frac{-j\pi(2m)}{}}e^{\frac{-j\pi(1)}{}}$$

$$X((4m+2)) = \sum_{n=0}^{\frac{N}{4}-1} (\ x(n) + x(n+\frac{N}{2})\ )e^{\frac{-j4\pi(2m)(n}{N}}e^{\frac{-j4\pi(1)(n}{N}} + \sum_{n=0}^{\frac{N}{4}-1} (\ x(n+\frac{N}{4}) + x(n+\frac{N}{2}+\frac{N}{4})\ )e^{\frac{-j4\pi(2m)(n)}{N}}e^{\frac{-j4\pi(1)(n)}{N}}(-1)$$

$$X((4m+2)) = \sum_{n=0}^{\frac{N}{4}-1} (\ x(n) + x(n+\frac{N}{2})\ )e^{\frac{-j4\pi(2m)(n)}{N}}e^{\frac{-j4\pi(1)(n)}{N}} - \sum_{n=0}^{\frac{N}{4}-1} (\ x(n+\frac{N}{4}) + x(n+\frac{N}{2}+\frac{N}{4})\ )e^{\frac{-j4\pi(2m)(n)}{N}}e^{\frac{-j4\pi(1)(n)}{N}}$$

$$X(4m+2) = \sum_{n=0}^{\frac{N}{4}-1} (\ x(n) + x(n+\frac{N}{2})\ )e^{\frac{-j4\pi n}{N}}e^{\frac{-j2\pi 4mn}{N}} - \sum_{n=0}^{\frac{N}{4}-1} (\ x(n+\frac{N}{4}) + x(n+\frac{N}{2}+\frac{N}{4})\ )e^{\frac{-j4\pi n}{N}}e^{\frac{-j2\pi 4mn}{N}}$$

$$X(2m+1) = \sum_{n=0}^{\frac{N}{2}-1} ((x(n) - x(n+\frac{N}{2}))e^{\frac{-j2\pi(1)n}{N}})e^{\frac{-j2\pi(2m)n}{N}}$$

$$X(2m+1) = \sum_{n=0}^{\frac{N}{4}-1} ((x(n) - x(n+\frac{N}{2}))e^{\frac{-j2\pi(1)n}{N}}e^{\frac{-j2\pi(2m)n}{N}} + \sum_{n=0}^{\frac{N}{4}-1} ((x(n+\frac{N}{4}) - x(n+\frac{N}{2}+\frac{N}{4}))e^{\frac{-j2\pi(1)(n+\frac{N}{4})}{N}}e^{\frac{-j2\pi(2m)(n+\frac{N}{4})}{N}}$$

$$X(2m+1) = \sum_{n=0}^{\frac{N}{4}-1} ((x(n) - x(n+\frac{N}{2}))e^{\frac{-j2\pi(1)n}{N}}e^{\frac{-j2\pi(2m)n}{N}} + \sum_{n=0}^{\frac{N}{4}-1} ((x(n+\frac{N}{4}) - x(n+\frac{N}{2}+\frac{N}{4}))e^{\frac{-j2\pi(1)(n)}{N}}e^{\frac{-j2\pi(1)(\frac{N}{4})}{N}}e^{\frac{-j2\pi(2m)(n)}{N}}e^{\frac{-j2\pi(2m)(\frac{N}{4})}{N}}$$

$$X(2m+1) = \sum_{n=0}^{\frac{N}{4}-1} ((x(n) - x(n+\frac{N}{2}))e^{\frac{-j2\pi n}{N}}e^{\frac{-j2\pi 2mn}{N}} + \sum_{n=0}^{\frac{N}{4}-1} ((x(n+\frac{N}{4}) - x(n+\frac{N}{2}+\frac{N}{4}))e^{\frac{-j2\pi n}{N}}e^{\frac{-j2\pi}{4}}e^{\frac{-j2\pi 2mn}{N}}e^{\frac{-j2\pi 2m}{4}}$$

$$X(2(2m)+1) = \sum_{n=0}^{\frac{N}{4}-1} ((x(n) - x(n+\frac{N}{2}))e^{\frac{-j2\pi n}{N}}e^{\frac{-j2\pi 2(2m)n}{N}} + \sum_{n=0}^{\frac{N}{4}-1} ((x(n+\frac{N}{4}) - x(n+\frac{N}{2}+\frac{N}{4}))e^{\frac{-j2\pi n}{N}}e^{\frac{-j2\pi}{4}}e^{\frac{-j2\pi 2(2m)n}{N}}e^{\frac{-j2\pi 2(2m)}{4}}$$

$$X(2(2m)+1) = \sum_{n=0}^{\frac{N}{4}-1} ((x(n) - x(n+\frac{N}{2}))e^{\frac{-j2\pi n}{N}}e^{\frac{-j2\pi 4mn}{N}} + \sum_{n=0}^{\frac{N}{4}-1} ((x(n+\frac{N}{4}) - x(n+\frac{N}{2}+\frac{N}{4}))e^{\frac{-j2\pi n}{N}}e^{\frac{-j2\pi}{4}}e^{\frac{-j2\pi 4mn}{N}}e^{\frac{-j2\pi 4m}{4}}$$

$$X(4m+1) = \sum_{n=0}^{\frac{N}{4}-1} ((x(n) - x(n+\frac{N}{2}))e^{\frac{-j2\pi n}{N}}e^{\frac{-j2\pi 4mn}{N}} + \sum_{n=0}^{\frac{N}{4}-1} ((x(n+\frac{N}{4}) - x(n+\frac{N}{2}+\frac{N}{4}))e^{\frac{-j2\pi n}{N}}e^{\frac{-j2\pi}{4}}e^{\frac{-j2\pi 4mn}{N}}e^{\frac{-j2\pi 4m}{4}}$$

$$X(4m+1) = \sum_{n=0}^{\frac{N}{4}-1} ((x(n) - x(n+\frac{N}{2}))e^{\frac{-j2\pi n}{N}}e^{\frac{-j2\pi 4mn}{N}} + j\sum_{n=0}^{\frac{N}{4}-1} ((x(n+\frac{N}{4}) - x(n+\frac{N}{2}+\frac{N}{4}))e^{\frac{-j2\pi n}{N}}e^{\frac{-j2\pi 4mn}{N}}e^{\frac{-j2\pi 4m}{4}}$$

$$X(4m+1) = \sum_{n=0}^{\frac{N}{4}-1} ((x(n) - x(n+\frac{N}{2}))e^{\frac{-j2\pi n}{N}}e^{\frac{-j2\pi 4mn}{N}} + j\sum_{n=0}^{\frac{N}{4}-1} ((x(n+\frac{N}{4}) - x(n+\frac{N}{2}+\frac{N}{4}))e^{\frac{-j2\pi n}{N}}e^{\frac{-j2\pi 4mn}{N}}$$

$$X(2m+1) = \sum_{n=0}^{\frac{N}{4}-1} ((x(n) - x(n+\frac{N}{2}))e^{\frac{-j2\pi n}{N}} e^{\frac{-j2\pi 2mn}{N}} + \sum_{n=0}^{\frac{N}{4}-1} ((x(n+\frac{N}{4}) - x(n+\frac{N}{2}+\frac{N}{4}))e^{\frac{-j2\pi n}{N}} e^{\frac{-j2\pi}{4}} e^{\frac{-j2\pi 2mn}{N}} e^{\frac{-j2\pi 2m}{4}}$$

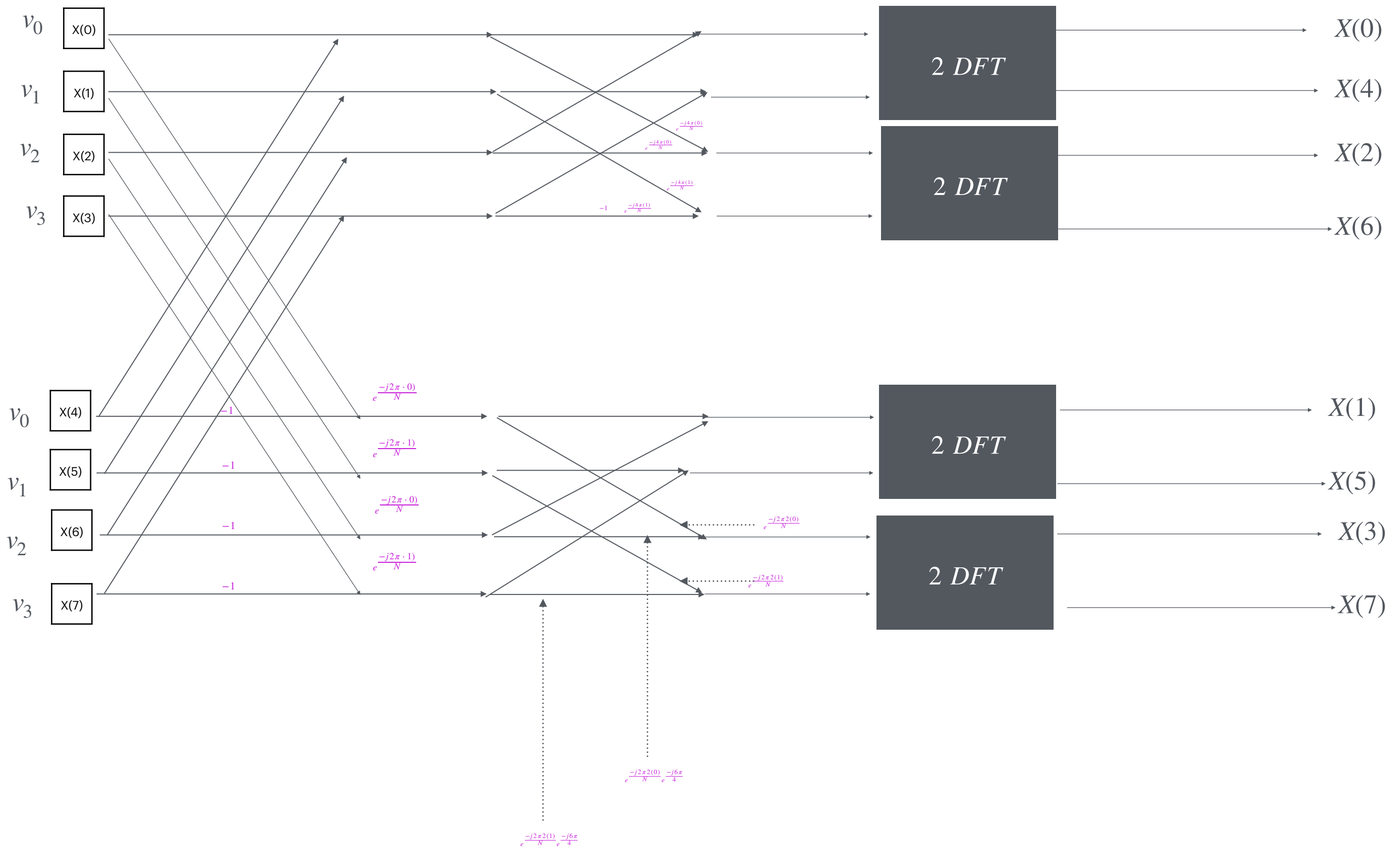
$$X(2(2m+1)+1) = \sum_{n=0}^{\frac{N}{4}-1} ((x(n) - x(n+\frac{N}{2}))e^{\frac{-j2\pi n}{N}} e^{\frac{-j2\pi 2(2m+1)n}{N}} + \sum_{n=0}^{\frac{N}{4}-1} ((x(n+\frac{N}{4}) - x(n+\frac{N}{2}+\frac{N}{4}))e^{\frac{-j2\pi n}{N}} e^{\frac{-j2\pi}{4}} e^{\frac{-j2\pi 2(2m+1)n}{N}} e^{\frac{-j2\pi 2(2m+1)}{4}}$$

$$X((4m+3)) = \sum_{n=0}^{\frac{N}{4}-1} ((x(n) - x(n+\frac{N}{2}))e^{\frac{-j2\pi n}{N}} e^{\frac{-j2\pi 2(1)n}{N}} e^{\frac{-j2\pi 2(2m)n}{N}} + \sum_{n=0}^{\frac{N}{4}-1} ((x(n+\frac{N}{4}) - x(n+\frac{N}{2}+\frac{N}{4}))e^{\frac{-j2\pi n}{N}} e^{\frac{-j2\pi 2n(1)}{N}} e^{\frac{-j2\pi}{4}} e^{\frac{-j2\pi 2(2m)}{4}} e^{\frac{-j2\pi 2(1)}{4}} e^{\frac{-j2\pi 2n(2m)}{N}}$$

$$X((4m+3)) = \sum_{n=0}^{\frac{N}{4}-1} ((x(n) - x(n+\frac{N}{2}))e^{\frac{-j2\pi n}{N}} e^{\frac{-j2\pi 2(1)n}{N}} e^{\frac{-j2\pi 2(2m)n}{N}} + \sum_{n=0}^{\frac{N}{4}-1} ((x(n+\frac{N}{4}) - x(n+\frac{N}{2}+\frac{N}{4}))e^{\frac{-j2\pi n}{N}} e^{\frac{-j2\pi 2n(1)}{N}} e^{\frac{-j2\pi(4m)}{4}} e^{\frac{-j2\pi 2(1)}{4}} e^{\frac{-j2\pi}{4}} e^{\frac{-j2\pi 2n(2m)}{N}}$$

$$X((4m+3)) = \sum_{n=0}^{\frac{N}{4}-1} ((x(n) - x(n+\frac{N}{2}))e^{\frac{-j2\pi n}{N}} e^{\frac{-j2\pi 2(1)n}{N}} e^{\frac{-j2\pi 2(2m)n}{N}} + \sum_{n=0}^{\frac{N}{4}-1} ((x(n+\frac{N}{4}) - x(n+\frac{N}{2}+\frac{N}{4}))e^{\frac{-j2\pi n}{N}} e^{\frac{-j2\pi 2n}{N}} e^{\frac{-j2\pi(4m)}{4}} e^{\frac{-j6\pi}{4}} e^{\frac{-j2\pi 2n(2m)}{N}}$$

$$X((4m+3)) = \sum_{n=0}^{\frac{N}{4}-1} ((x(n) - x(n+\frac{N}{2}))e^{\frac{-j2\pi n}{N}} e^{\frac{-j2\pi 2n}{N}} e^{\frac{-j2\pi 2(2m)n}{N}} + \sum_{n=0}^{\frac{N}{4}-1} ((x(n+\frac{N}{4}) - x(n+\frac{N}{2}+\frac{N}{4}))e^{\frac{-j2\pi n}{N}} e^{\frac{-j2\pi 2n}{N}} e^{\frac{-j6\pi}{4}} e^{\frac{-j2\pi 2n(2m)}{N}}$$


v_0
v_1
v_2
v_3
X(0)
X(1)
X(2)
X(3)
v_0
v_1
v_2
v_3
X(4)
X(5)
X(6)
X(7)
-1
$e^{\frac{-j2\pi \cdot 0)}{N}}$
$e^{\frac{-j2\pi \cdot 1)}{N}}$
$e^{\frac{-j2\pi \cdot 0)}{N}}$
$e^{\frac{-j2\pi \cdot 1)}{N}}$
-1
-1
-1
-1
$e^{\frac{-j4\pi(0)}{N}}$
$e^{\frac{-j8\pi(0)}{N}}$
$e^{\frac{-j4\pi(1)}{N}}$
-1
$e^{\frac{-j4\pi(1)}{N}}$
$e^{\frac{-j2\pi 2(0)}{N}}$
$e^{\frac{-j2\pi 2(1)}{N}}$
$e^{\frac{-j2\pi 2(0)}{N}} e^{\frac{-j6\pi}{4}}$
$e^{\frac{-j2\pi 2(1)}{N}} e^{\frac{-j6\pi}{4}}$
2 DFT
2 DFT
2 DFT
2 DFT
$X(0)$
$X(4)$
$X(2)$
$X(6)$
$X(1)$
$X(5)$
$X(3)$
$X(7)$